Pietro Gasparetto

Engineering portfolio

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Linkedin

Education

POLITECNICO MILANO 1863

Bachelor degree – Politecnico di Milano
Mechanical engineering
Sep 2019 – Sep 2024
Final grade: 99/110

Master degree – Politecnico di Milano
Mechanical engineering:
Automotive and Motorsport Engineering
Sep 2024 – Oct 2026
Current avg grade: 29,1/30

Formula Student

Suspension and Drivetrain – Team Member
Sep 2019 – Aug 2021

Head of Suspension and Drivetrain
Sep 2021 – Aug 2022

Technical Director – Mechanical
Sep 2022 – Sep 2023

Drivetrain and *Powertrain* – Team Member
Sep 2023 – Jun 2024

IAC & A2RL

POLIMOVE

Vehicle Dynamics and Control Engineer
Feb 2026 – Present

Formula Student

Two-stage gearbox for outboard electric motors



Role:

- Mechanical designer

Objectives:

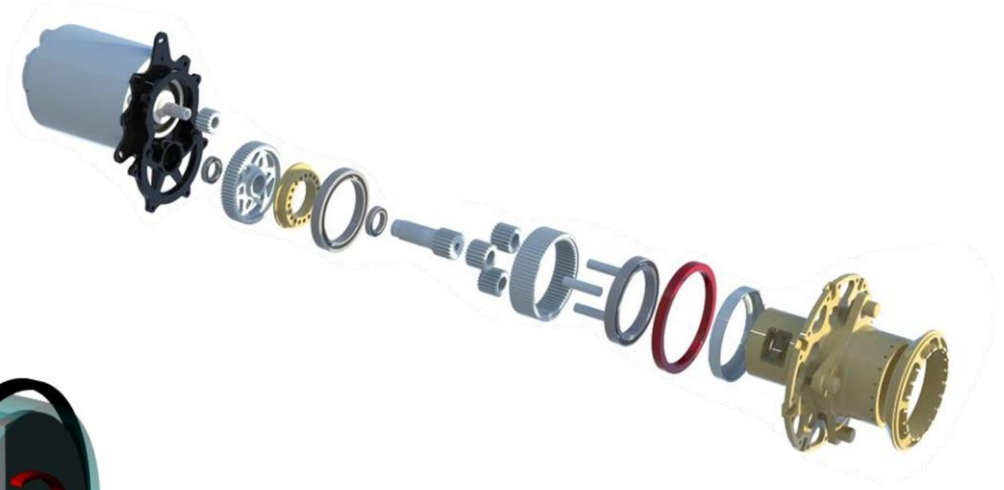
- Minimize mass and volume of the gearbox
- Ensure reliability and efficiency

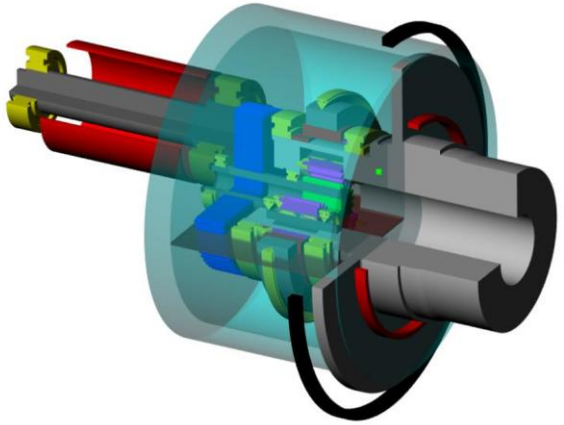
Methods:

- Matlab genetic algorithm optimization
- CAD modelling
- KISSsoft and KISSsys calculations

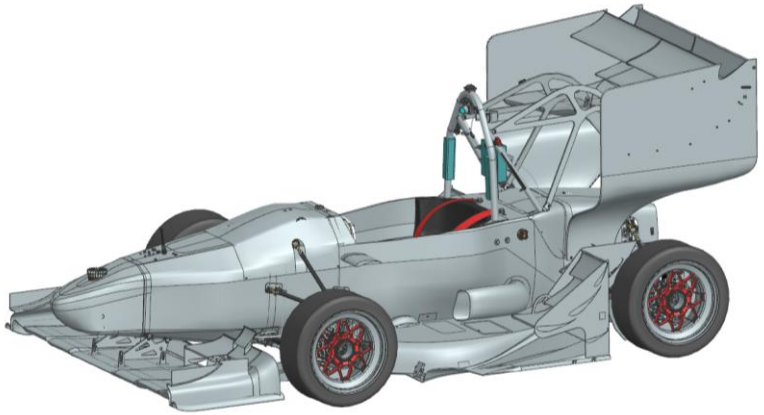
Results:

- Successfully designed and realized the gearbox which performed without any reliability concerns





Technical Director – Mechanical



Role:

- Technical and organizational coordination between mechanical departments and vehicle dynamics
- Technical interface with electronic/electrical departments
- Contacts with sponsors and suppliers

Objectives:

- Obtain the best possible results with the available time and resources

Results:

- P1 autox DV – FSG
- P3 overall EV – FSA
- Early development of next year car, accelerating the subsequent design cycle



Electric Motor Design & Test Bench



Role:

- Mechanical development of three phase IPMSM

Objectives:

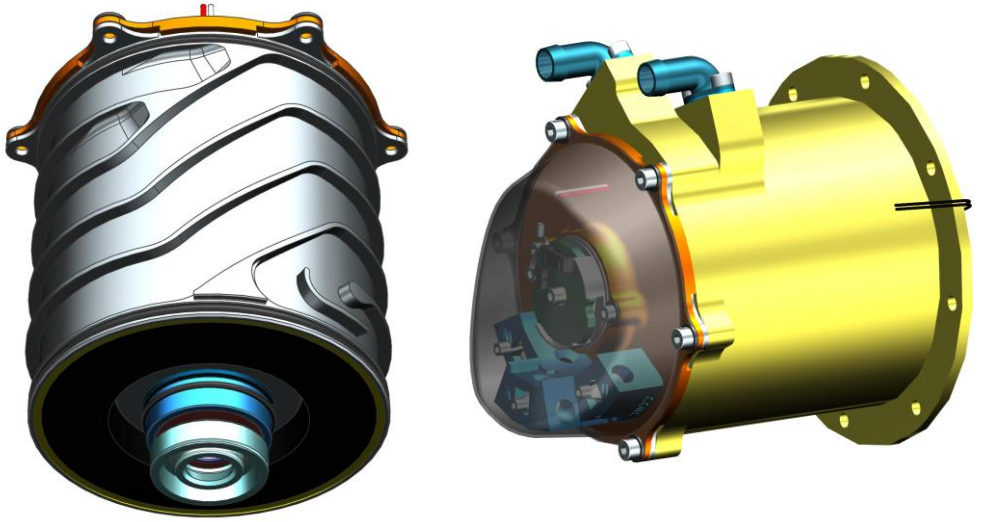
- Provide minimum mass and optimal package integration
- Ensure motor reliability

Solutions:

- Two-pieces titanium hollow shaft
- In-wheel motor design with stator integrated cooling jacket

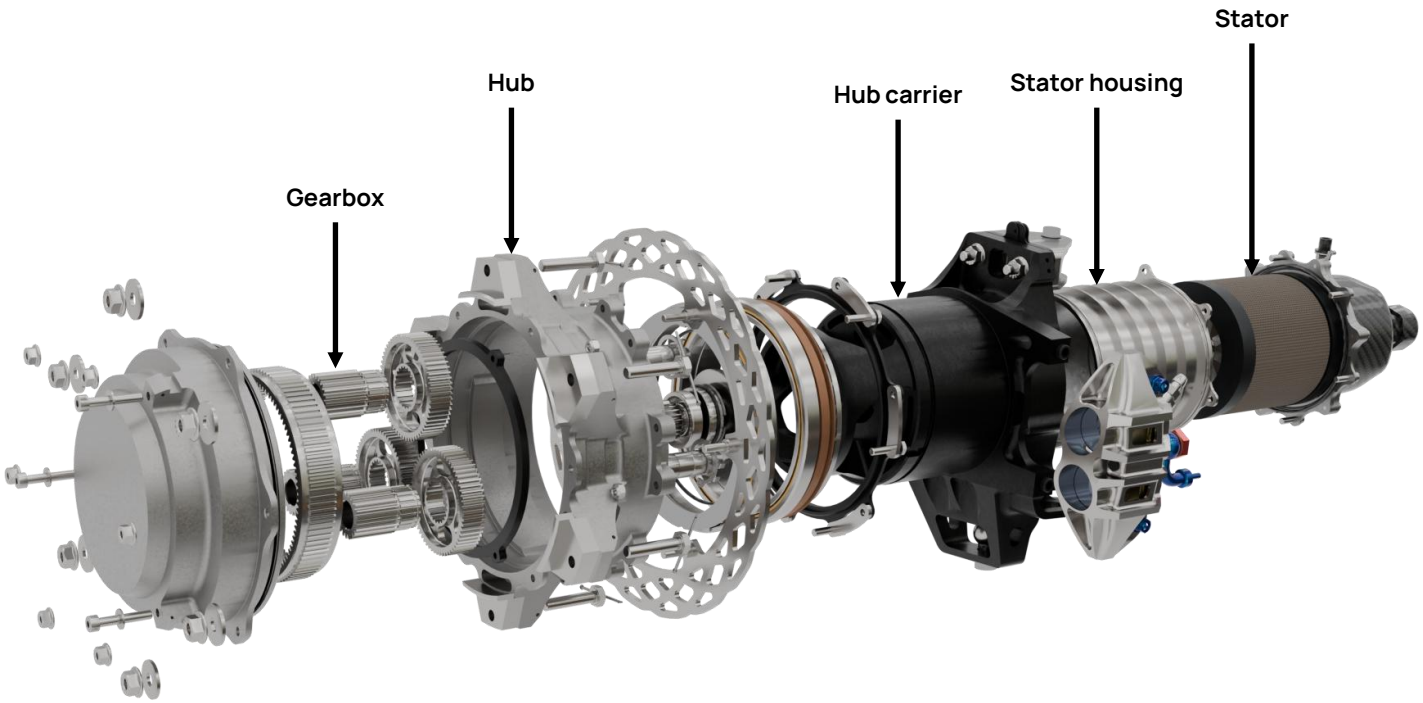
Results:

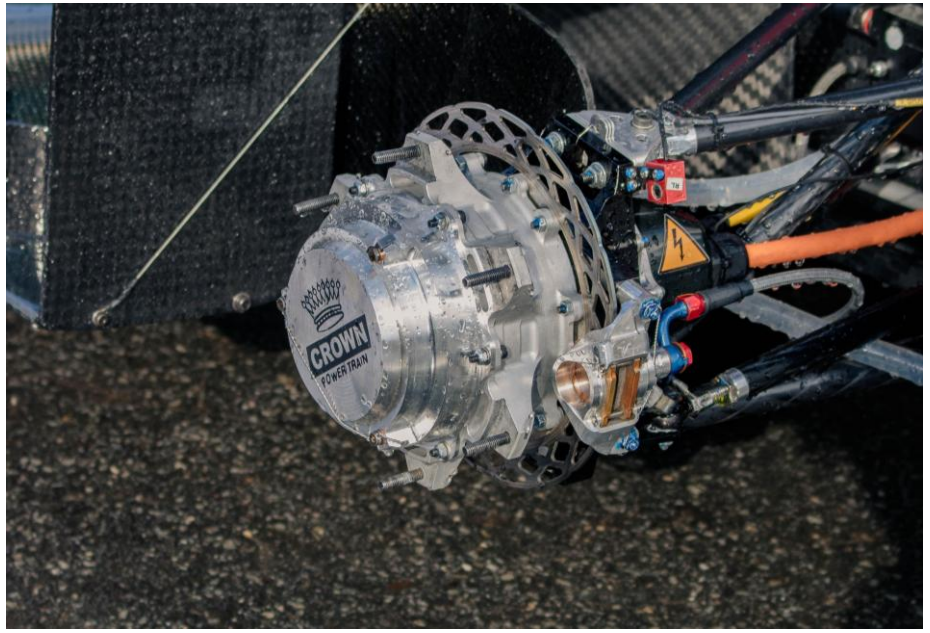
- Team's first semi-custom electric motor
- Mass: $3{,}55kg \rightarrow 3{,}05\ kg^*$
- Torque output: $21Nm \rightarrow 29Nm^*$
- **Garrett Powertrain Innovation Award**



* compared to previous year commercial solution

Electric Motor Design & Test Bench

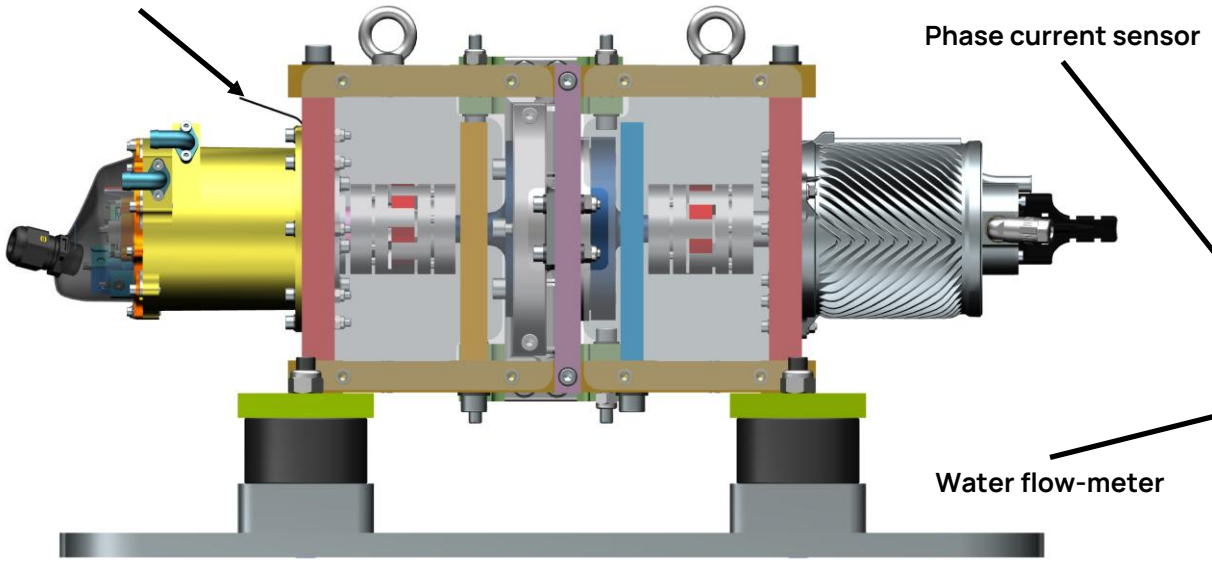


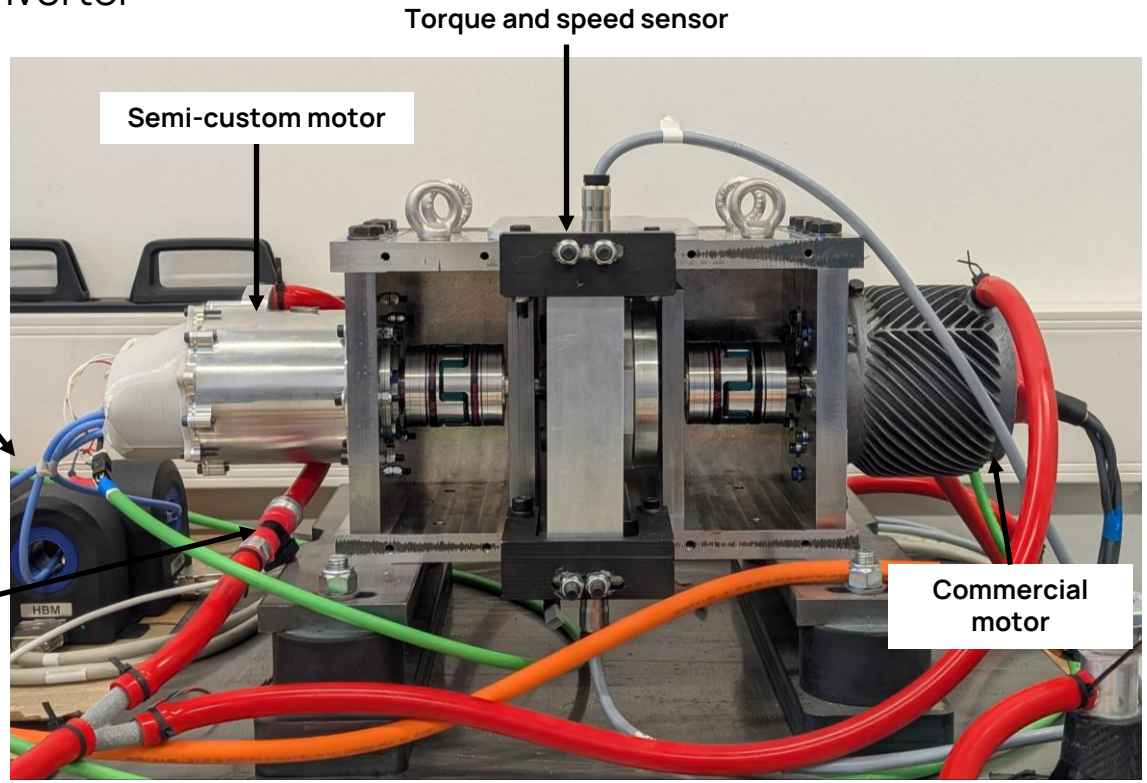


Electric Motor Design & Test Bench

Objectives:

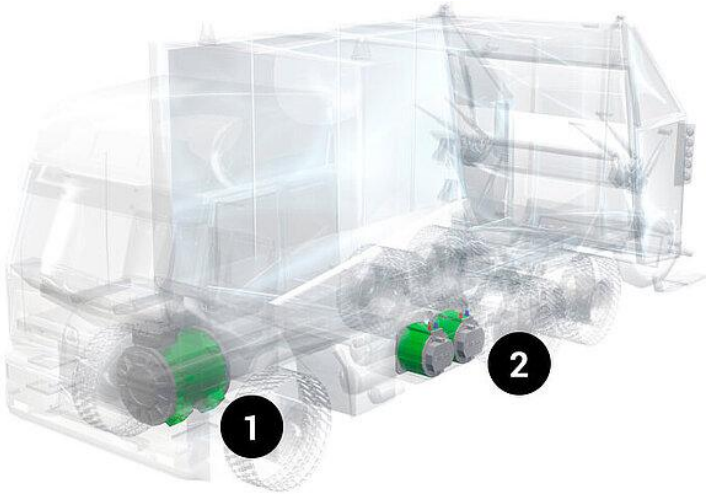
- Prove reliability over the most extreme working range
- Extensive characterization of efficiency maps of motor and inverter





Academic projects

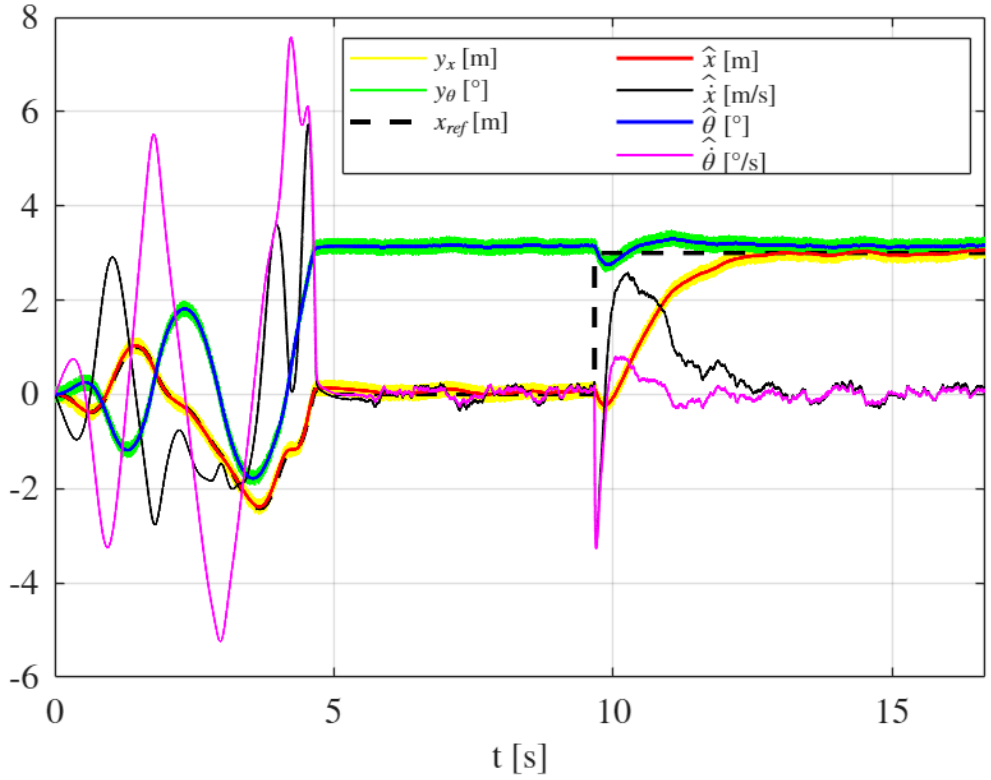
Hybridization of a heavy duty vehicle

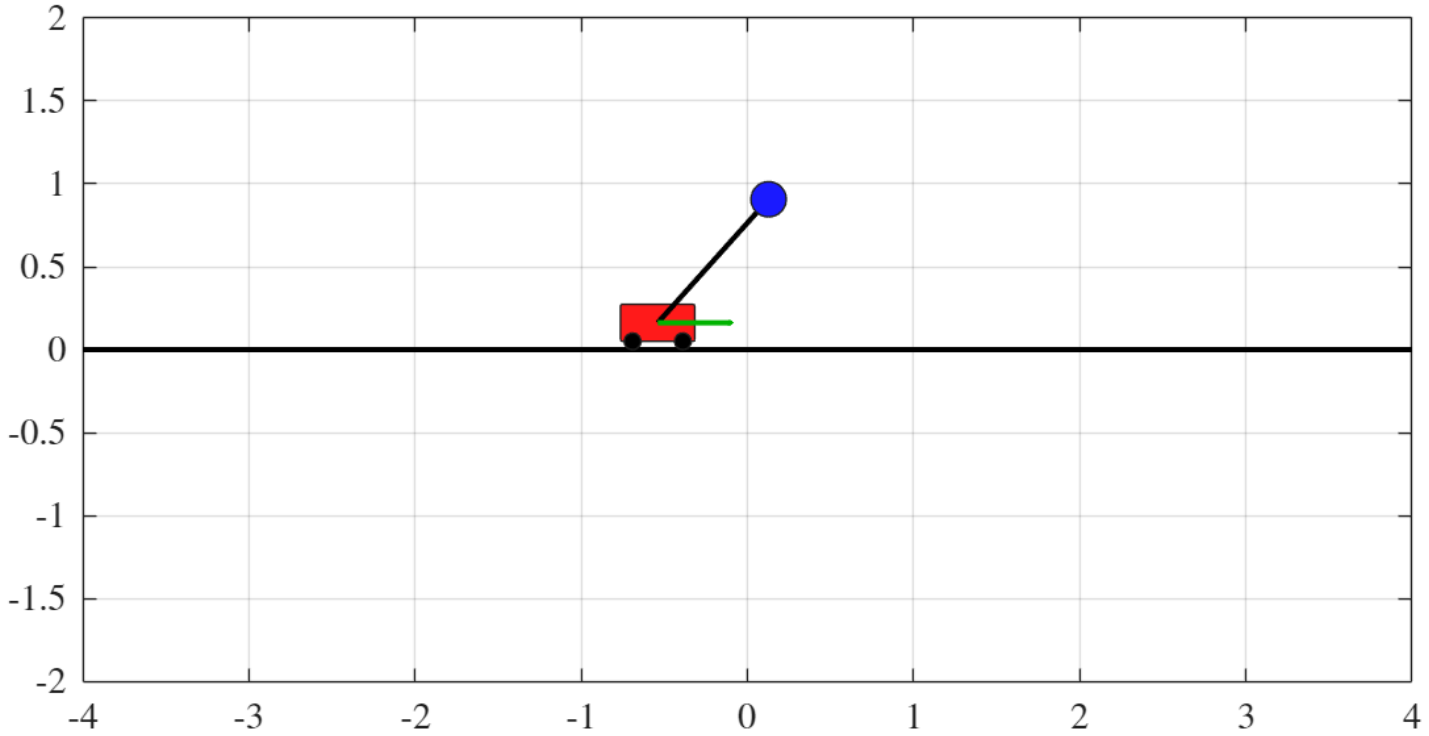


- Sizing of battery pack and electric motors
- Development of high and low level control logic, including torque split between ICE and EM
- Modelling through Matlab & Simulink of:
  - Longitudinal dynamics
  - Motors, battery, transmission, termo-electro-mechanical interaction of components
  - State machine for different driving modes
  - Control logics and safety protections

Optimal control: trajectory, feedback, estimation

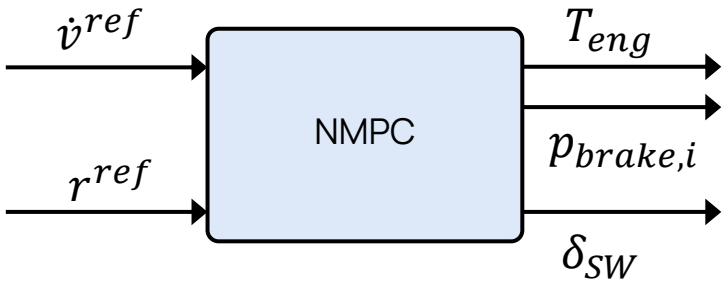
- Offline computation of optimal trajectory, according to a defined cost function
- Nonlinear state estimation, accounting for process and sensor noise
- Online optimal control

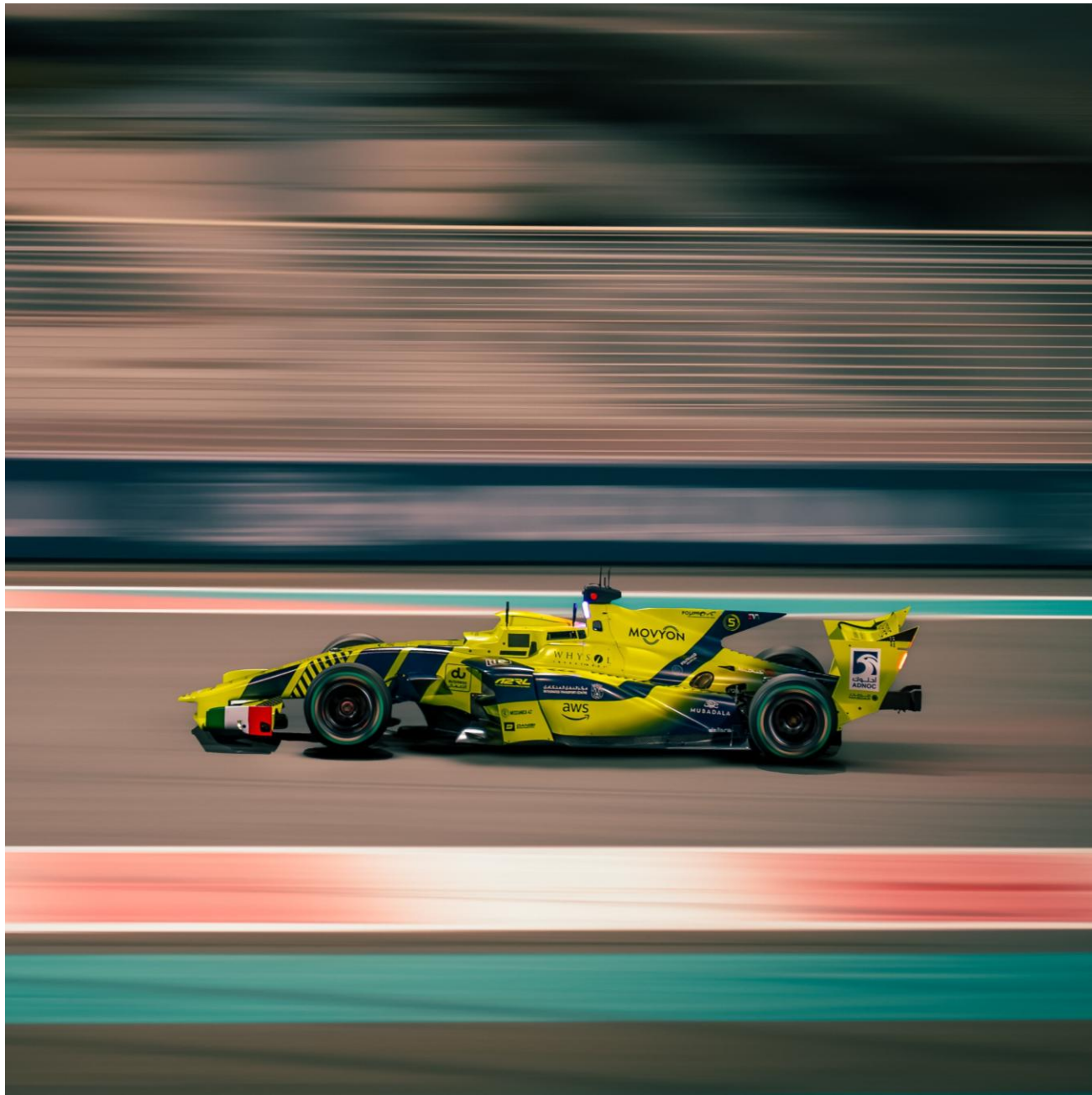




NMPC for coupled longitudinal-lateral control of high performance driverless racing vehicle

- Non-linear Model Predictive Controller based on double-track vehicle model
- Tracking of longitudinal and lateral references over a 0.5 s prediction horizon discretized at 100 Hz
  - Actuation via RWD engine torque, 4-wheel independent brakes, and front steering
- Real Time Iteration (RTI) scheme implemented via CasADi in a Matlab/Simulink environment





Contatti

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